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Sec: 08

1a) We know,

$$I = \int r^2 \, dm$$

$$dm = \frac{M}{L} dr$$

Given, range,

$x = -\frac{L}{2}$ to $x = \frac{L}{2}$

$$I = \int_{-L/2}^{L/2} x^2 \left(\frac{M}{L}\right) dx$$

$$= \frac{M}{L} \int_{-L/2}^{L/2} x^2 \, dx$$

$$= \frac{M}{L} \left[\frac{x^3}{3}\right]_{-L/2}^{L/2}$$

$$= \frac{M}{L} \times \left(\frac{(L/2)^3}{3} - \frac{(-L/2)^3}{3}\right)$$

$$= \frac{1}{12} \times \frac{ML^3}{L}$$

$$= \frac{1}{12} ML^2 \text{ (Ans)}$$

b) $I = I$ (center of mass) $+ M \times$ perpendicular height

$$= \frac{1}{12} ML^2 + M\left(\frac{1}{2}L\right)^2$$

$$= \frac{1}{12} ML^2 + M\left(\frac{1}{4} ML^2\right)$$

$$= \frac{1}{3} ML^2$$

c) Since the rod is melted and formed into a sphere,

Volume of sphere = volume of rod

$$\Rightarrow \frac{4}{3}\pi r^3 = 0.001 \times L$$

$$\Rightarrow r^3 = \frac{3 \times 0.001 \times L}{4\pi}$$

$$= \sqrt[3]{\frac{0.003L}{4\pi}} \text{ (Ans)}$$

We know,

the rotational inertia of the solid sphere.

$$I = \frac{2}{5} Mr^2$$

$$= \frac{2}{5} \times 1 \times \sqrt[3]{\frac{0.003L}{4\pi}}$$

$$= \frac{2}{5}\left(\frac{0.003}{4\pi}\right)^{2/3} \text{ (Ans)}$$

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2a) Area of Coin = $2\pi r\, dr$

$$\sigma = \frac{M}{\pi R^2}$$

$$dm = \frac{M}{\pi R^2} \times 2\pi r\, dr$$

$$= \frac{2Mr\,dr}{R^2}$$

$$I = \int_0^R r^2\, dm$$

$$= \int_0^R r^2 \times \frac{2Mr}{R^2}\, dr$$

$$= \frac{2M}{R^2} \int_0^R r^3\, dr$$

$$= \frac{2M}{R^2} \left[\frac{r^4}{4}\right]_0^R$$

$$= \frac{2M}{R^2} \times \frac{R^4}{4}$$

$$= \frac{MR^2}{2} \text{ (Ans)}$$

b) Inertia of the center perpendicular axis, $I_z = \frac{1}{2} MR^2$

Applying perpendicular axis theorem,

$$I_z = I_x + I_y$$

$$\Rightarrow I_z = 2 I_x$$

$$\Rightarrow I_x = \frac{1}{2 \times 2} MR^2 = \frac{1}{4} MR^2$$

∴ the moment of ~~intere~~ inertia about the diameter of the coin

$$= \frac{1}{4} MR^2$$

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3a) $I = \int r^2 \, dm$

$$\frac{dm}{dr} = \frac{M}{L}$$

$$I = \int_0^L r^2 \frac{M}{L} dr$$

$$= \frac{M}{L} \int_0^L r^2 \, dr$$

$$= \frac{M}{L} \left[\frac{r^3}{3}\right]_0^L$$

$$= \frac{1}{3} \times M \times \frac{L^3}{L}$$

$$I = \frac{L^2 M}{3} \quad \text{(Ans)}$$

b) Given,

$M = 2Kg$

$L = 1m$

$g = 9.8 \ m/s^2$

Initial GPE $= Mg\left(\frac{L}{2}\right)$

$KE = \frac{1}{2} I\omega^2$

$$\Rightarrow \frac{MLg}{2} = \frac{1}{2} I\omega^2$$

$$\Rightarrow \frac{MgL}{2} = \frac{1}{2} \times \frac{1}{3} ML^2\omega^2$$

$$\frac{g}{2} = \frac{1}{6} \omega^2 L$$

$$\frac{3g}{L} = \omega^2$$

$$\therefore \omega = \sqrt{\frac{3g}{L}}$$

$$= \sqrt{\frac{3 \times 9.8}{1}}$$

$$= 5.42 \ rad/s \text{ (Ans)}$$

c) $V = \omega r$

$= \omega \times L$

$= 5.42 \times 1$

$= 5.42 \ m/s$ (Ans)

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